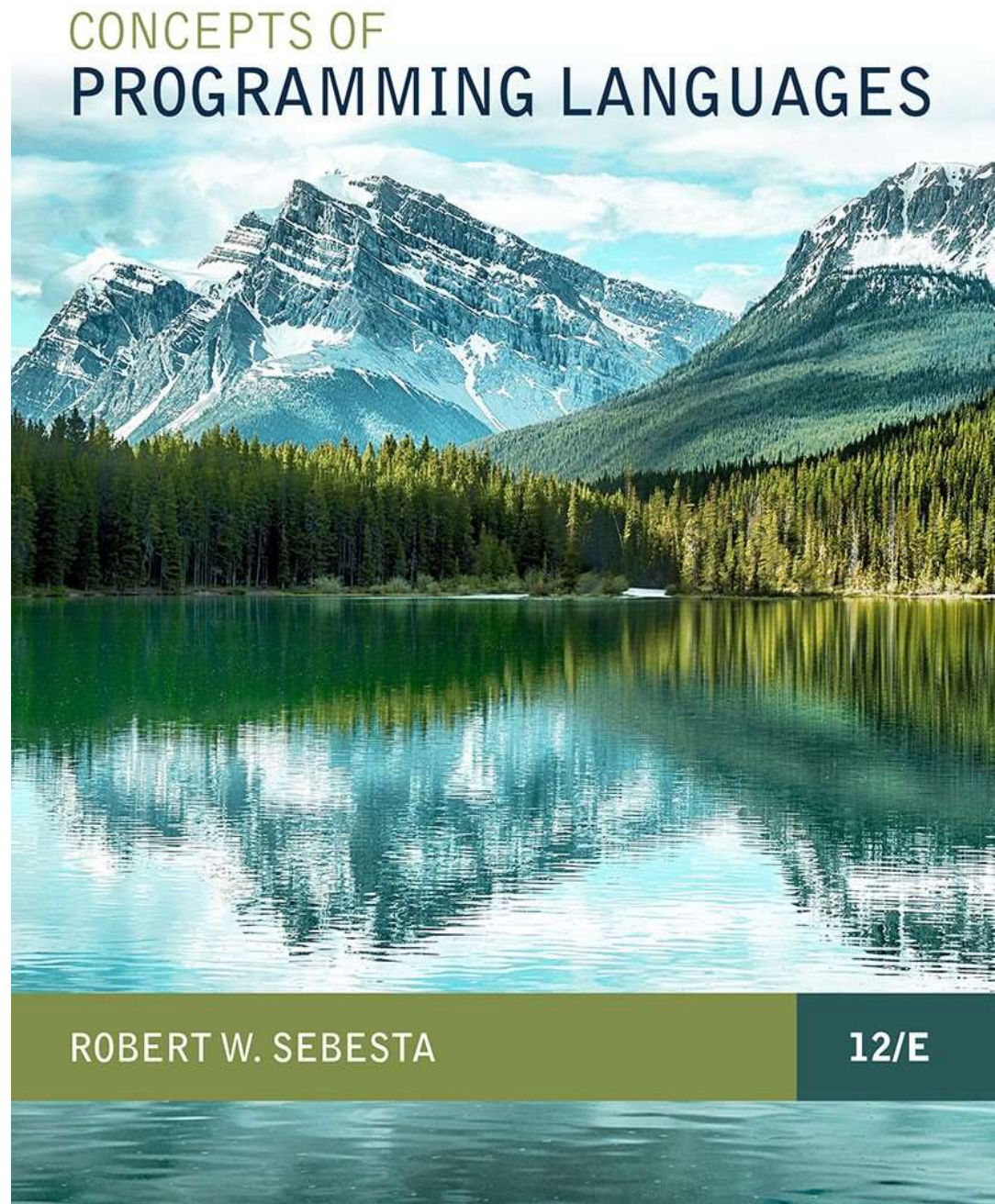


Chapter 1

Preliminaries

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Reasons for Studying Concepts of Programming Languages

- Increased ability to express ideas
- Improved background for choosing appropriate languages
- Increased ability to learn new languages
- Better understanding of significance of implementation
- Better use of languages that are already known
- Overall advancement of computing

Programming Domains

- **Scientific** applications
 - Large numbers of floating point computations; use of arrays
 - Fortran
- **Business** applications
 - Produce reports, use decimal numbers and characters
 - COBOL
- **Artificial intelligence**
 - Symbols rather than numbers manipulated; use of linked lists
 - LISP
- **Systems programming**
 - Need efficiency because of continuous use
 - C
- **Web Software**
 - Eclectic collection of languages: markup (e.g., HTML), scripting (e.g., PHP), general-purpose (e.g., Java)

Language Evaluation Criteria

- **Readability:** the ease with which programs can be read and understood
- **Writability:** the ease with which a language can be used to create programs
- **Reliability:** conformance to specifications (i.e., performs to its specifications)
- **Cost:** the ultimate total cost

Evaluation Criteria: Readability

- Overall simplicity
 - A manageable set of features and constructs
 - Minimal feature multiplicity
 - Minimal operator overloading
- Orthogonality
 - A relatively small set of primitive constructs can be combined in a relatively small number of ways
 - Every possible combination is legal
- Data types
 - Adequate predefined data types
- Syntax considerations
 - Identifier forms: flexible composition
 - Special words and methods of forming compound statements
 - Form and meaning: self-descriptive constructs, meaningful keywords

Evaluation Criteria: Writability

- Simplicity and orthogonality
 - Few constructs, a small number of primitives, a small set of rules for combining them
- Support for abstraction
 - The ability to define and use complex structures or operations in ways that allow details to be ignored
- Expressivity
 - A set of relatively convenient ways of specifying operations
 - Strength and number of operators and predefined functions

Evaluation Criteria: Reliability

- Type checking
 - Testing for type errors
- Exception handling
 - Intercept run-time errors and take corrective measures
- Aliasing
 - Presence of two or more distinct referencing methods for the same memory location
- Readability and writability
 - A language that does not support “natural” ways of expressing an algorithm will require the use of “unnatural” approaches, and hence reduced reliability

Evaluation Criteria: Cost

- Training programmers to use the language
- Writing programs (closeness to particular applications)
- Executing programs
- Reliability: poor reliability leads to high costs
- Maintaining programs

Evaluation Criteria: Others

- **Portability**
 - The ease with which programs can be moved from one implementation to another
- **Generality**
 - The applicability to a wide range of applications
- **Well-definedness**
 - The completeness and precision of the language's official definition

Influences on Language Design

- **Computer Architecture**

- Languages are developed around the prevalent computer architecture, known as the *von Neumann* architecture

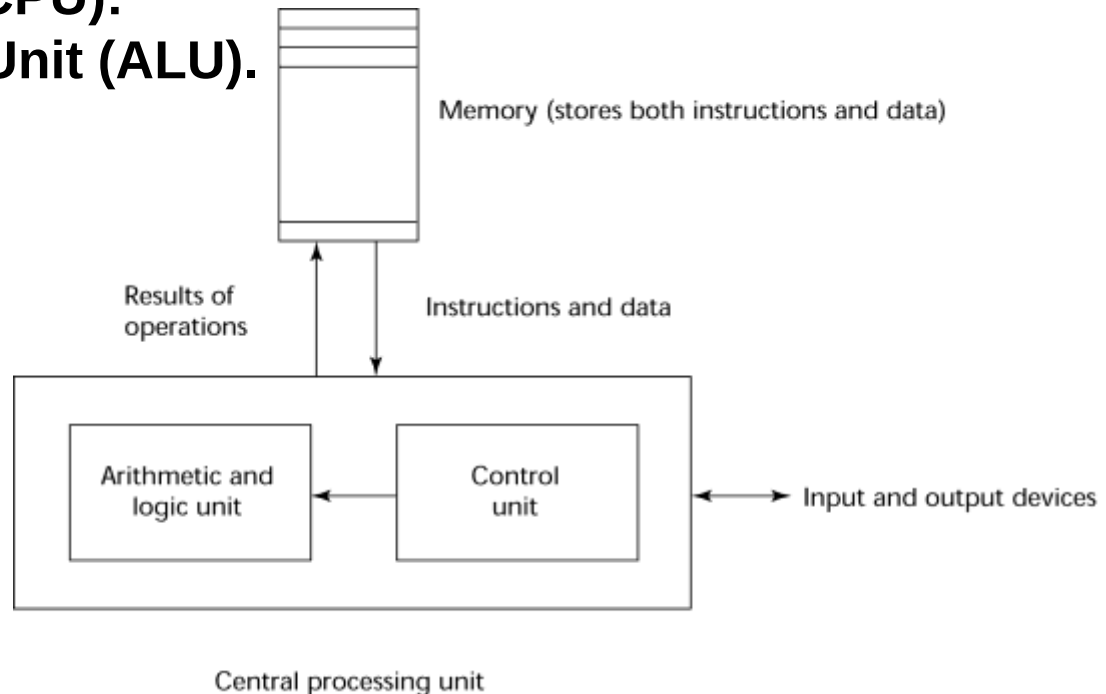
- **Program Design Methodologies/Paradigms**

- New software development methodologies (e.g., object-oriented software development) led to new programming paradigms and by extension, new programming languages

The von Neumann Architecture

Components of von Neumann Machine:

- **Input Devices.**
- **Central Processing Unit (CPU):**
 - **Arithmetic and Logic Unit (ALU).**
 - **Control Unit (CU).**
- **Output Devices**
- **Memory**



Computer Architecture Influence

- Well-known computer architecture: Von Neumann
- Imperative languages, most dominant, because of von Neumann computers
 - Data and programs stored in memory
 - Memory is separate from CPU
 - Instructions and data are piped from memory to CPU
 - Basis for imperative languages
 - Variables model memory cells
 - Assignment statements model piping
 - Iteration is efficient

Software–Hardware Dependency

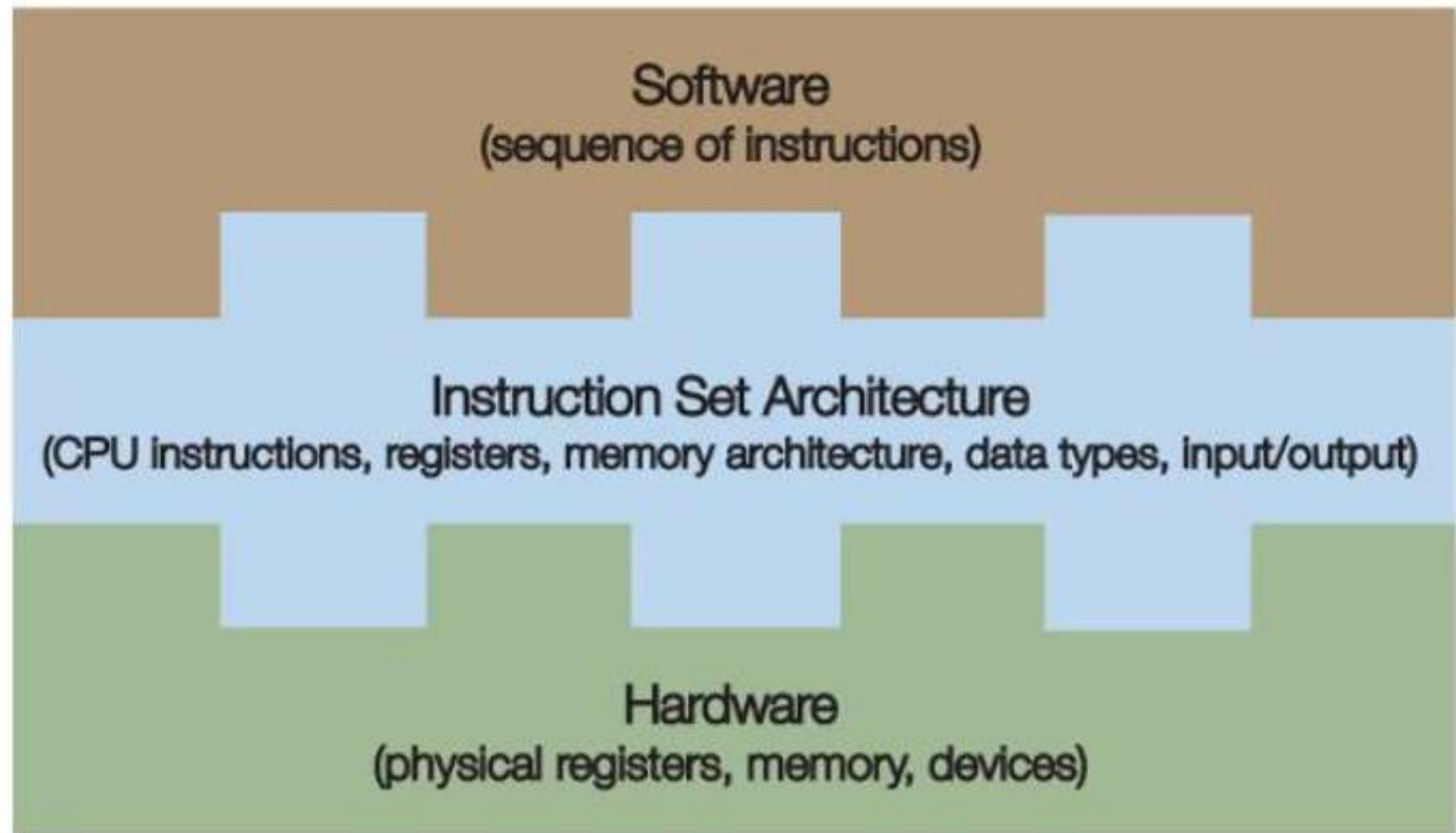


Figure 1.3 Instruction set architecture

Translation Pipeline

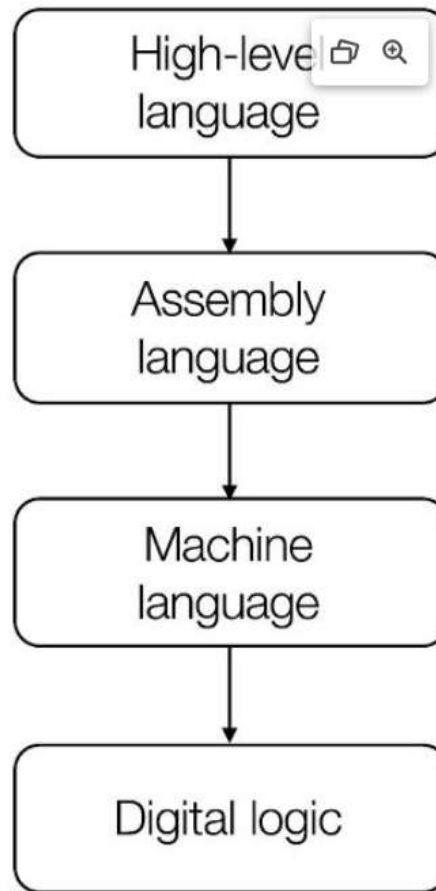


Figure 1.2 Translation pipeline

The von Neumann Architecture

- Fetch–execute–cycle (on a von Neumann architecture computer)

```
initialize the program counter
```

```
repeat forever
```

```
    fetch the instruction pointed by the counter
```

```
    increment the counter
```

```
    decode the instruction
```

```
    execute the instruction
```

```
end repeat
```

Programming Methodologies Influences

- 1950s and early 1960s: Simple applications; worry about machine efficiency
- Late 1960s: People efficiency became important; readability, better control structures
 - structured programming
 - top-down design and step-wise refinement
- Late 1970s: Process-oriented to data-oriented
 - data abstraction
- Middle 1980s: Object-oriented programming
 - Data abstraction + inheritance + polymorphism

Language Categories

- Imperative
 - Central features are variables, assignment statements, and iteration
 - Include languages that support object-oriented programming
 - Include scripting languages
 - Include the visual languages
 - Examples: C, Java, Perl, JavaScript, Visual BASIC .NET, C++
- Functional
 - Main means of making computations is by applying functions to given parameters
 - Examples: LISP, Scheme, ML, F#
- Logic
 - Rule-based (rules are specified in no particular order)
 - Example: Prolog
- Markup/programming hybrid
 - Markup languages extended to support some programming
 - Examples: JSTL, XSLT

Language Design Trade-Offs

- **Reliability vs. cost of execution**
 - Example: Java demands all references to array elements be checked for proper indexing, which leads to increased execution costs
- **Readability vs. writability**

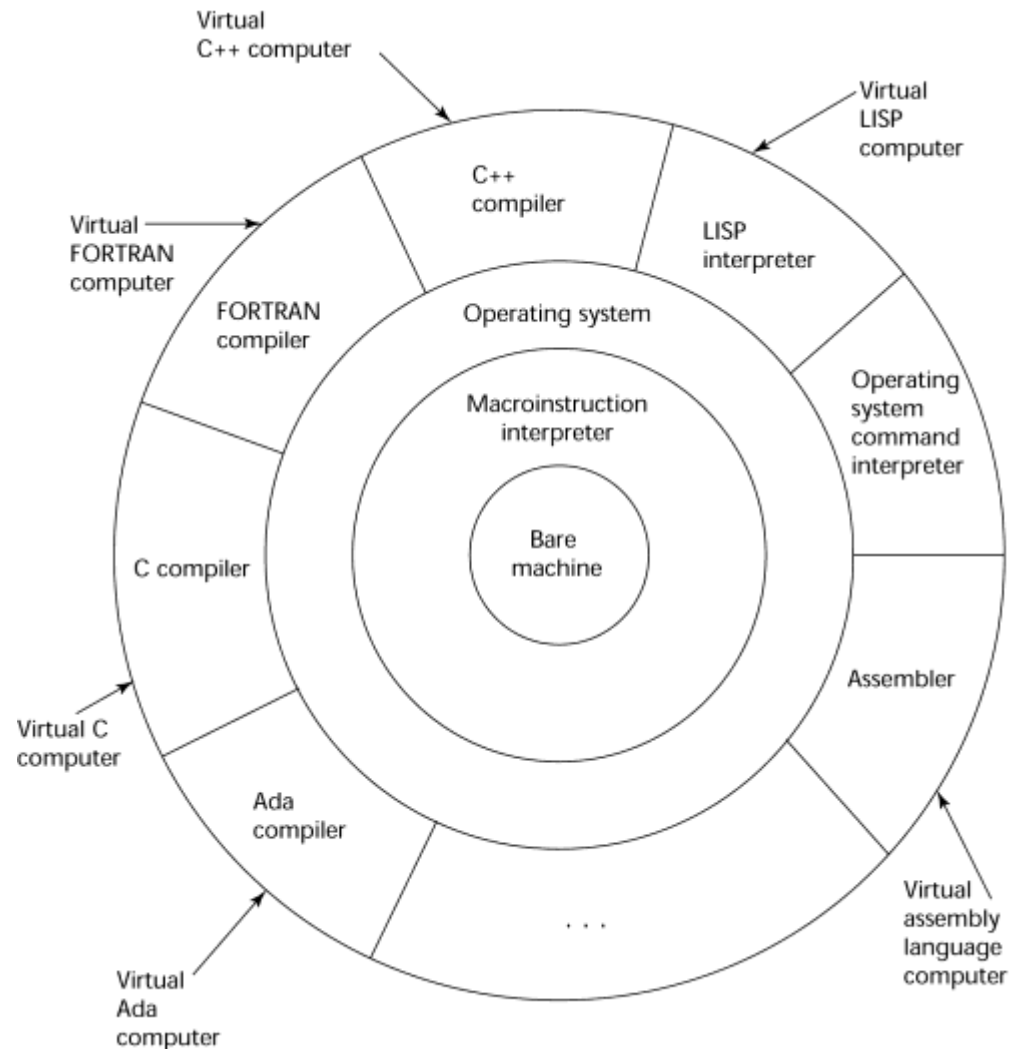
Example: APL provides many powerful operators (and a large number of new symbols), allowing complex computations to be written in a compact program but at the cost of poor readability
- **Writability (flexibility) vs. reliability**
 - Example: C++ pointers are powerful and very flexible but are unreliable

Implementation Methods

- **Compilation**
 - Programs are translated into machine language; includes JIT systems
 - Use: Large commercial applications
- **Pure Interpretation**
 - Programs are interpreted by another program known as an interpreter
 - Use: Small programs or when efficiency is not an issue
- **Hybrid Implementation Systems**
 - A compromise between compilers and pure interpreters
 - Use: Small and medium systems when efficiency is not the first concern

Layered View of Computer

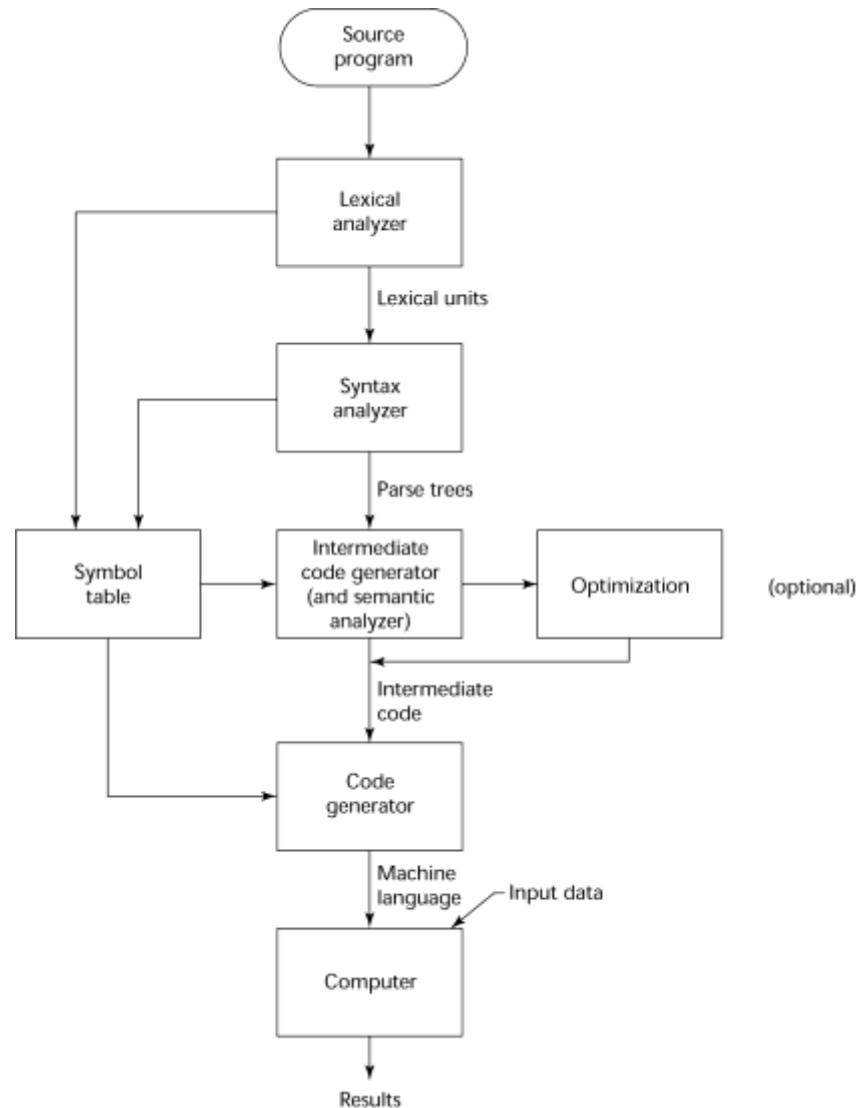
The operating system and language implementation are layered over machine interface of a computer



Compilation

- Translate high-level program (source language) into machine code (machine language)
- Slow translation, fast execution
- Compilation process has several phases:
 - lexical analysis: converts characters in the source program into lexical units
 - syntax analysis: transforms lexical units into *parse trees* which represent the syntactic structure of program
 - Semantics analysis: generate intermediate code
 - code generation: machine code is generated

The Compilation Process



Additional Compilation Terminologies

- **Load module** (executable image): the user and system code together
- **Linking and loading**: the process of collecting system program units and linking them to a user program

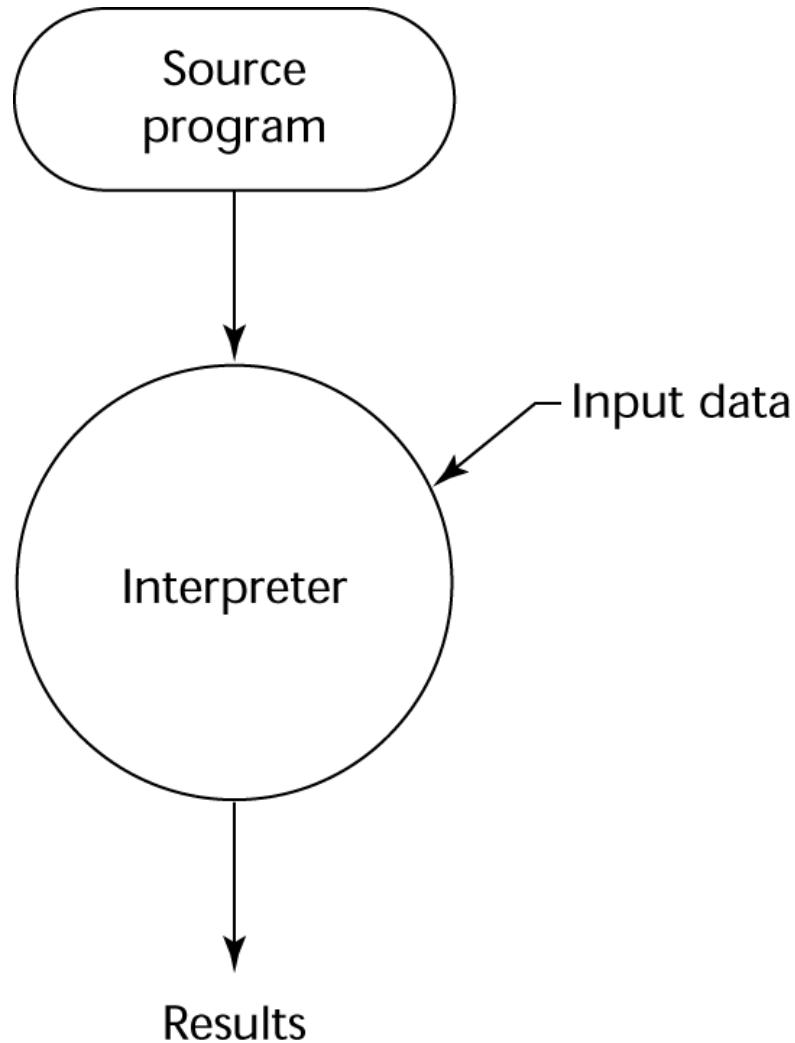
Von Neumann Bottleneck

- Connection speed between a computer's memory and its processor determines the speed of a computer
- Program instructions often can be executed much faster than the speed of the connection; the connection speed thus results in a *bottleneck*
- Known as the *von Neumann bottleneck*; it is the primary limiting factor in the speed of computers

Pure Interpretation

- No translation
- Easier implementation of programs (run-time errors can easily and immediately be displayed)
- Slower execution (10 to 100 times slower than compiled programs)
- Often requires more space
- Now rare for traditional high-level languages
- Significant comeback with some Web scripting languages (e.g., JavaScript, PHP)

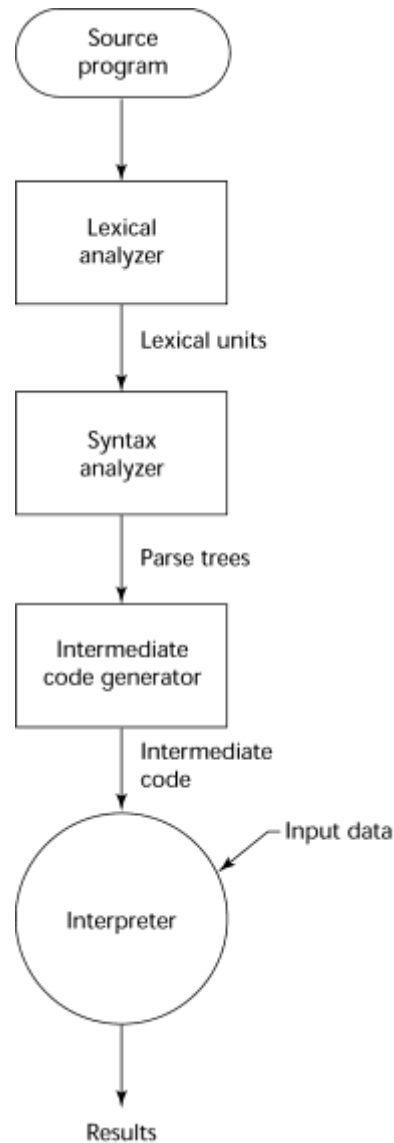
Pure Interpretation Process



Hybrid Implementation Systems

- A compromise between compilers and pure interpreters
- A high-level language program is translated to an intermediate language that allows easy interpretation
- Faster than pure interpretation
- Examples
 - Perl programs are partially compiled to detect errors before interpretation
 - Initial implementations of Java were hybrid; the intermediate form, *byte code*, provides portability to any machine that has a byte code interpreter and a run-time system (together, these are called *Java Virtual Machine*)

Hybrid Implementation Process



Just-in-Time Implementation Systems

- Initially translate programs to an intermediate language
- Then compile the intermediate language of the subprograms into machine code when they are called
- Machine code version is kept for subsequent calls
- JIT systems are widely used for Java programs
- .NET languages are implemented with a JIT system
- In essence, JIT systems are delayed compilers

Preprocessors

- Preprocessor macros (instructions) are commonly used to specify that code from another file is to be included
- A preprocessor processes a program immediately before the program is compiled to expand embedded preprocessor macros
- A well-known example: C preprocessor
 - expands `#include`, `#define`, and similar macros

Programming Environments

- A collection of tools used in software development
- UNIX
 - An older operating system and tool collection
 - Nowadays often used through a GUI (e.g., CDE, KDE, or GNOME) that runs on top of UNIX
- Microsoft Visual Studio.NET
 - A large, complex visual environment
- Used to build Web applications and non-Web applications in any .NET language
- NetBeans
 - Related to Visual Studio .NET, except for applications in Java

Summary

- The study of programming languages is valuable for a number of reasons:
 - Increase our capacity to use different constructs
 - Enable us to choose languages more intelligently
 - Makes learning new languages easier
- Most important criteria for evaluating programming languages include:
 - Readability, writability, reliability, cost
- Major influences on language design have been machine architecture and software development methodologies
- The major methods of implementing programming languages are: compilation, pure interpretation, and hybrid implementation